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Printed Pages – 7

328873 (28)

BE (8th Semester)

Examination, Nov.-Dec., 2013

Branch : Et & T

INFORMATION THEORY & CODING (PE-III)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : (i) Attempt all questions.

(ii) Part (a) of each question is compulsory.

(iii) Attempt any two parts from (b), (c), (d) of each question.

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(iv) Any printing mistake or missing data may be assumed suitably and mention clearly.

(v) Standard Notation have usual meaning.

1. (a) What is average information or Entropy. 2

(b) A discrete source transmits message x_1, x_2, and x_3 with the probability 0.3, 0.4 and 0.3. The source is connected to the channel given in Figure 1.b. Calculate all the entropies. 7

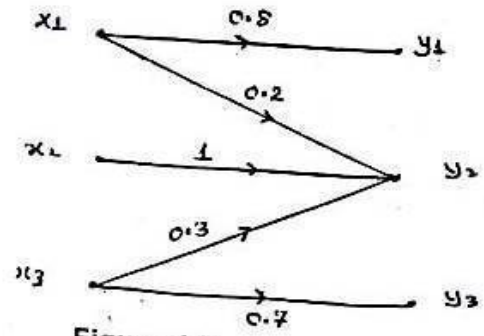

Figure 1.b

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(c) Find the mutual information and channel capacity of the channel shown in figure 1.c. Given $P(x_1) = 0.6$, $P(x_2) = 0.4$. 7

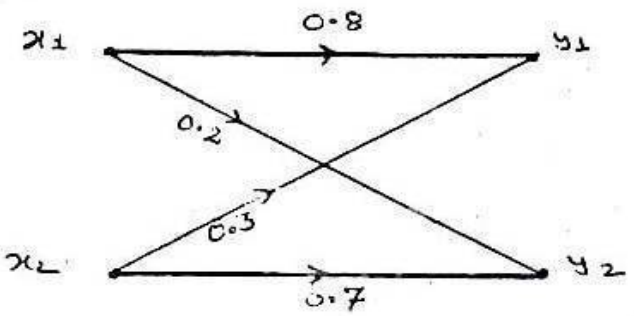

Figure 1.c.

(d) Draw and explain the model of digital communication system. 7

Q. 2. (a) What is the difference between systematic codes and non systematic codes. 2

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(b) The parity check matrix of a particular (7, 4) linear block code is given by : 7

$$[H] = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

(i) Find the generator matrix (G).

(ii) List all the code vectors.

(iii) What is the minimum distance between code vector ?

(iv) How many Error can be detected ? How many error can be corrected ?

(c) (i) Explain information capacity theorem. 5

(ii) How many error can be detected and connected if minimum distance is 3 2

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(d) For a systematic linear block code, the three parity check digits C_4, C_5 and C_6 are given by : 7

$C_4 = d_1 \oplus d_2 \oplus d_3$

$C_5 = d_1 \oplus d_2$

$C_6 = d_2 \oplus d_3$

(i) Construct generator matrix.

(ii) Construct code generated by this matrix.

(iii) Determine error correcting capability.

(iv) Prepare a suitable decoding table.

(v) Decode the received words 101100 and 000110.

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Q. 3. (a) What is meant by Image Compression ?

(b) Construct a Huffman code for the sour symbol, whose probabilities are 0.4, 0 0.12, 0.08, 0.08, 0.08, 0.04.

(c) Explain CCITT group 3 1-D compressio Mention its advantage and disadvantages.

(d) Explain the fixed length and variable len coding. What is the importance of varia length coding.

Q. 4. (a) What is compression technique ? What t of technique used in winzip compression

(b) Draw and explain encoder and decoder MPEG compression technique.

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(c) Explain how audio file compressed. 7

(d) Describe requirement of full motion video compression. 7

Q. 5. (a) Define transposition cipher with advantages. 2

(b) What are two different uses of public key cryptography related to key distribution ? 7

(c) Briefly explain Diffie-Heliman key exchange. 7

(d) Draw and explain data encryption standard and its algorithm in details. Also state what are the limitation of data encryption standard process ? 7

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